# HubSpot Integration

## 1. Overview

### 1.1 Description

**HubSpot Integration** connects **HubSpot Meetings + CRM** with **Newo Platform**.

It currently supports:

- checking appointment availability from a HubSpot meeting link
- creating bookings with contact lookup / contact creation in HubSpot CRM
- cancelling existing meetings by exact booking datetime
- existing customer lookup by phone/email with prompt injection
- storing upcoming bookings in persona data
- optional outbound conversation note sync after phone sessions
- optional default canvas seeding for booking and cancellation scenarios

This integration is designed for **service businesses and sales teams** that use a single HubSpot meeting link and want an AI agent to handle appointment availability, booking, and cancellation through chat or voice.

### 1.2 Use Cases

- When a customer asks about available times -> the agent checks HubSpot meeting-link availability and returns free slots
- When a customer wants to book an appointment -> the agent finds or creates a HubSpot contact and submits the booking
- When a customer wants to cancel an appointment -> the agent finds the matching HubSpot meeting by start time and deletes it
- When the platform detects a caller's phone number -> the integration looks up the existing HubSpot contact and injects customer context
- When a future meeting exists for that contact -> the integration stores it in persona `bookings`
- When an outbound phone session ends -> the integration can write a HubSpot note with transcript/summary

### 1.3 Supported Features

| Feature | Supported | Notes |
|---|---|---|
| Check Availability | ✅ | Uses HubSpot meeting-link booking endpoint |
| Book Appointment | ✅ | Searches contact by email, creates contact if missing, then books |
| Cancel Appointment | ✅ | Finds meeting by exact start datetime, then deletes it |
| Existing Client Detection | ✅ | Searches HubSpot contacts by phone/email |
| Existing Booking Storage | ✅ | Saves future meetings into persona `bookings` |
| Silent Availability Preload | ✅ | `session_started` can trigger a silent availability check |
| Outbound Conversation Note Sync | ✅ | Only for `Outgoing Phone` sessions and when enabled |

| | | |
|---|---|---|
| Canvas Auto Setup | ✅ | Seeds default booking/cancellation scenarios when enabled |
| Multi-Calendar / Multi-Slug | ❌ | Single `hubspot_slug_name` |
| Native Reschedule Flow | ❌ | Use cancel + rebook flow |
| Deal / Pipeline Automation | ❌ | Not implemented in current agent flows |

## 2. Requirements

Before setup:

- active **HubSpot** account
- **HubSpot Meetings** enabled
- one working **meeting link slug**
- either:
  - one **OAuth authorization code** from the Newo-provided HubSpot auth link
  - or one **HubSpot Service Key** (PAT-style token) with the scopes your booking flows require
- correct business timezone

Current setup supports **two authentication modes**:

- `OAuth`
- `Token`

## 3. How to Setup

### 3.1 Step 1 — Choose Authentication Method

Set `hubspot_auth_type` to one of:

- `OAuth`
- `Token`

### 3.2 OAuth Setup

If you use `OAuth`:

1. Open the authorization link from the `hubspot_oauth_code` attribute description in Newo.
2. Approve access in HubSpot.
3. Copy the returned OAuth code.
4. Paste it into `hubspot_oauth_code`.

During setup, the integration uses that code when creating the HubSpot connector and then also triggers the connector's `authorization_code` action.

### 3.3 Token Setup

1. Open HubSpot.
2. Go to **Settings > Integrations > Service Keys**.
3. Create a new key with the CRM / Meetings permissions your flow needs.
4. Copy the generated token.

### 3.4 Step 2 — Find the Meeting Link Slug

Example:

- full link: `https://meetings-eu1.hubspot.com/test-meetings-page`
- slug: `test-meetings-page`

Put only the last path segment into `hubspot_slug_name`.

### 3.5 Step 3 — Configure Attributes in Newo

Open **Newo -> Projects -> Builder / Attributes** and set:

| Attribute | Required | Description |
|---|---|---|
| `hubspot_auth_type` | ✅ | `OAuth` or `Token` |
| `hubspot_oauth_code` | Required for OAuth | One-time OAuth code copied from the HubSpot authorization page |
| `hubspot_service_key` | Required for Token | HubSpot Service Key / PAT |
| `hubspot_slug_name` | ✅ | HubSpot meeting link slug |
| `hubspot_base_api_url` | ❌ | Defaults to `https://api.hubapi.com` |
| `hubspot_timezone` | ❌ | Defaults to project business timezone |
| `hubspot_duration` | ❌ | Slot length in minutes, default `30` |
| `hubspot_show_for_days` | ❌ | Search window in days, default `1` |
| `hubspot_slots_available` | ❌ | Enable / disable availability flow |
| `hubspot_booking_available` | ❌ | Enable / disable booking flow |
| `hubspot_cancelation_available` | ❌ | Enable / disable cancellation flow |
| `hubspot_sync_conversation` | ❌ | Enable / disable outbound phone note sync |
| `hubspot_setup_scenarios` | ❌ | Auto-add default booking/cancellation scenarios to canvas |

Hidden/system attributes are managed by setup automatically, for example:

- `hubspot_override_agent_attributes`
- `setup_persona_id`

### 3.6 Step 4 — Publish

On publish, the current setup logic:

- syncs project attributes and tool-gating attributes
- injects booking and availability payload schemas
- ensures both `api` and `http` connector entries exist for `hubspot_connector`
- creates / reuses the setup persona
- validates the HubSpot connection with: `GET /crm/v3/objects/contacts?limit=1`

Operational note:

- `OAuth` requests use the `api` connector path
- `Token` requests use the `http` connector path with `Authorization: Bearer <hubspot_service_key>`

If `hubspot_setup_scenarios = True`, the integration also participates in workflow-builder canvas setup through `CanvasSetupFlow`.

## 4. How It Works

### 4.1 Flows

| Flow | Purpose |
|---|---|
| `SetupFlow` | Attribute sync, tool gating, setup persona creation, connection validation |
| `CanvasSetupFlow` | Auto-seeds booking/cancellation canvas items when enabled |
| `UtilsFlow` | Normalizes outbound HubSpot requests and routes connector results |
| `AvailabilityFlow` | Checks meeting-link availability and returns free slots |
| `BookingFlow` | Searches/creates contact and books the meeting link |
| `CancellationFlow` | Finds the meeting by start time and deletes it |
| `ExistingClientFlow` | Finds HubSpot contact by phone/email and stores future meetings |
| `ConversationFlow` | Writes outbound phone conversation notes into HubSpot |
| `MetadataFlow` | Returns integration version |

### 4.2 Triggers

| Trigger | Event | Flow |
|---|---|---|
| Check Availability | `get_available_slots` | `AvailabilityFlow` |
| Book Appointment | `book_slot` | `BookingFlow` |
| Cancel Appointment | `convo_agent_cancel_booking` | `CancellationFlow` |
| Existing Client Lookup | `define_user_phone_number` | `ExistingClientFlow` |
| Silent Availability | `session_started` | `AvailabilityFlow` |
| Setup | `project_publish_finish` | `SetupFlow` |
| Canvas Rebuild | `gm_workflow_builder_canvas_built` | `CanvasSetupFlow` |
| Outbound Session Sync | `session_ended` | `ConversationFlow` |

### 4.3 Main API Actions

| Action | Method | Endpoint |
|---|---|---|
| Validate Connection | `GET` | `/crm/v3/objects/contacts?limit=1` |

| | | |
|---|---|---|
| Search Contact | `POST` | `/crm/v3/objects/contacts/search` |
| Create Contact | `POST` | `/crm/v3/objects/contacts` |
| Get Availability | `GET` | `/scheduler/v3/meetings/meeting-links/book/{slug}?timezone={tz}` |
| Book Meeting | `POST` | `/scheduler/v3/meetings/meeting-links/book?timezone={tz}` |
| Search Meetings | `POST` | `/crm/v3/objects/meetings/search` |
| Delete Meeting | `DELETE` | `/crm/v3/objects/meetings/{id}` |
| Create Note | `POST` | `/crm/v3/objects/notes` |

### 4.4 Overall Sequence

```
sequenceDiagram
    participant U as User
    participant A as ConvoAgent
    participant I as HubSpotAgent
    participant API as HubSpot API

    Note over I: Setup
    A->>I: project_publish_finish
    I->>API: GET /crm/v3/objects/contacts?limit=1
    API-->>I: connection validated

    Note over U,API: Existing Client
    A->>I: define_user_phone_number
    I->>API: POST /crm/v3/objects/contacts/search
    API-->>I: contact result
    I->>A: set_prompt_custom_section(ExistingCustomerInfo)
    I->>API: POST /crm/v3/objects/meetings/search
    API-->>I: future meetings

    Note over U,API: Availability
    U->>A: asks for slots
    A->>I: get_available_slots
    I->>API: GET /scheduler/v3/meetings/meeting-links/book/{slug}
    API-->>I: busy time payload
    I->>I: build free slots
    I->>A: result_success(availability)

    Note over U,API: Booking
    U->>A: confirms datetime
    A->>I: book_slot
    I->>API: POST /crm/v3/objects/contacts/search
    alt contact missing
        I->>API: POST /crm/v3/objects/contacts
        API-->>I: contact id
    end
    I->>API: POST /scheduler/v3/meetings/meeting-links/book
```

```
    API-->>I: booking result
    I->>A: result_success or result_error

    Note over U,API: Cancellation
    U->>A: asks to cancel
    A->>I: convo_agent_cancel_booking
    I->>API: POST /crm/v3/objects/meetings/search
    API-->>I: meeting id
    I->>API: DELETE /crm/v3/objects/meetings/{id}
    API-->>I: cancellation result
    I->>A: result_success or result_error
```

### 4.5 AvailabilityFlow

```
flowchart TD
    E["get_available_slots / session_started"] --> G{"hubspot_slots_available?"}
    G -->|"No"| SKIP["Return"]
    G -->|"Yes"| PREP["Build search window from date + hubspot_show_for_days"]
    PREP --> API["GET meeting-link booking endpoint"]
    API --> OK{"HTTP success?"}
    OK -->|"No"| ERR["ExecuteResultError -> result_error"]
    OK -->|"Yes"| BUSY["Read allUsersBusyTimes"]
    BUSY --> CALC["Generate free slots using hubspot_duration"]
    CALC --> OUT["result_success with availability[]"]
```

Current implementation detail:

- availability is generated from `allUsersBusyTimes`
- the flow does **not** use `linkAvailabilityByDuration` as the authoritative slot list

### 4.6 BookingFlow

```
flowchart TD
    E["book_slot"] --> G{"hubspot_booking_available?"}
    G -->|"No"| SKIP["Return"]
    G -->|"Yes"| SEARCH["POST contacts/search by email"]
    SEARCH --> FOUND{"Contact found?"}
    FOUND -->|"Yes"| BOOK["POST booking endpoint"]
    FOUND -->|"No"| CREATE["POST contacts"]
    CREATE --> BOOK
    BOOK --> OK{"calendarEventId present?"}
    OK -->|"Yes"| OUT["result_success"]
    OK -->|"No"| ERR["ExecuteResultError -> result_error"]
```

Important runtime details:

- contact search is by **email**
- booking uses exact `bookingDateTime`
- booking request body includes: `slug`, `firstName`, `lastName`, `email`, `startTime`, `duration`, `timezone`

### 4.7 CancellationFlow

```
flowchart TD
    E["convo_agent_cancel_booking"] --> G{"hubspot_cancelation_available?"}
    G -->|"No"| SKIP["Return"]
    G -->|"Yes"| UTC["Convert local datetime to UTC"]
    UTC --> FIND["POST meetings/search by hs_meeting_start_time"]
    FIND --> FOUND{"Meeting id found?"}
    FOUND -->|"No"| ERR["ExecuteResultError -> result_error"]
    FOUND -->|"Yes"| DEL["DELETE meeting by id"]
    DEL --> OUT["result_success / result_error"]
```

Important runtime detail:

- cancellation is driven by exact `datetime`
- the flow does **not** cancel by booking URL or contact association

### 4.8 ExistingClientFlow

```
flowchart TD
    E["define_user_phone_number"] --> SEARCH["POST contacts/search by phone OR
email"]
    SEARCH --> FOUND{"Contact found?"}
    FOUND -->|"No"| DONE["Return"]
    FOUND -->|"Yes"| INJECT["set_prompt_custom_section ExistingCustomerInfo"]
    INJECT --> APPTS["POST meetings/search by associations.contact"]
    APPTS --> FILTER["Keep only future meetings"]
    FILTER --> SAVE["Store into persona bookings"]
```

Runtime outputs:

- custom prompt section: `ExistingCustomerInfo role="context"`
- persona storage: `bookings`

### 4.9 ConversationFlow

This flow is only for outbound phone-note sync.

It runs on `session_ended` and:

- checks `hubspot_sync_conversation`
- only continues for channel `Outgoing Phone`
- resolves the contact by `contact_id`, phone, or email
- creates a HubSpot note with summary/transcript data

## 5. Canvas Setup

`CanvasSetupFlow` exists and auto-runs on:

- `gm_workflow_builder_canvas_built`

Current behavior:

- `SetupLibrarySkill` seeds custom `hubspot_*` intents and scenarios into the library
- `SetupCanvasSkill` currently inserts common appointment/cancellation library items into the active canvas
- auto-setup is controlled by visible attribute: `hubspot_setup_scenarios`

# 6. Testing

## 6.1 Conversation Testing

Recommended live checks:

1. publish the integration
2. verify setup validation succeeds
3. ask for availability on a concrete date/time
4. book a new customer with email + phone
5. cancel by exact appointment datetime
6. test existing-client lookup by known phone number

## 6.2 Webhook Test Events

The integration also exposes test hooks for direct flow testing:

| Event | Flow | Purpose |
|---|---|---|
| `availability_wh` | `AvailabilityFlow` | Trigger availability test payload |
| `book_wh` | `BookingFlow` | Trigger booking test payload |
| `cancel_wh` | `CancellationFlow` | Trigger cancellation test payload |

## 6.3 What to Verify

- `SetupFlow` : token validation request succeeds
- `CanvasSetupFlow` : default items are added when `hubspot_setup_scenarios=True`
- `AvailabilityFlow` : returned slots match the current meeting-link behavior
- `BookingFlow` : contact search/create works and booking endpoint accepts the chosen time
- `CancellationFlow` : meeting search finds the correct meeting and delete succeeds
- `ExistingClientFlow` : `ExistingCustomerInfo` appears and future bookings are stored

## 6.4 Troubleshooting

If setup fails:

- if using `OAuth` , confirm `hubspot_oauth_code` was pasted correctly and connector authorization completed
- if using `Token` , confirm `hubspot_service_key` is valid
- confirm the OAuth app / token has the scopes needed by your booking flows
- confirm `hubspot_base_api_url` is correct

If availability works but booking fails:

- verify the chosen time is still valid for the meeting link
- verify timezone alignment between `hubspot_timezone` , user request, and HubSpot meeting settings

- note that the current implementation computes free slots from `allUsersBusyTimes`; HubSpot may still reject a time if it does not fit the meeting link's live availability settings

If existing-client lookup does nothing:

- confirm the incoming phone number/email matches a HubSpot contact
- confirm the contact has future meetings associated to it

If outbound note sync does nothing:

- confirm `hubspot_sync_conversation=True`
- confirm the session channel is exactly `Outgoing Phone`

## 7. FAQ

**Q: What authentication method does the current setup use?**
A: The current setup supports both `OAuth` and `Token` through `hubspot_auth_type`.

**Q: What does OAuth mode require from the user?**
A: In the current implementation, the user opens the auth link from `hubspot_oauth_code`, approves HubSpot access, copies the returned code, and pastes it into `hubspot_oauth_code`.

**Q: Does the current runtime use `api` or `http` routing?**
A: `OAuth` uses the `api` path. `Token` uses the `http` path with a bearer token header.

**Q: How are contacts matched during booking?**
A: Booking first searches HubSpot contacts by **email**. If no contact is found, a new contact is created.

**Q: How does cancellation identify the meeting to delete?**
A: It converts the requested local datetime to UTC, searches meetings by exact `hs_meeting_start_time`, then deletes the found meeting ID.

**Q: What customer context is injected into the prompt?**
A: `ExistingClientFlow` injects `ExistingCustomerInfo role="context"` and stores future meetings in persona `bookings`.

**Q: Why can availability say a slot is open but booking still fail?**
A: The current availability logic derives free slots from `allUsersBusyTimes`. HubSpot booking still validates the submitted `startTime` against its own live meeting-link availability settings.

**Q: Can I disable the default canvas scenarios?**
A: Yes. Set `hubspot_setup_scenarios` to `False`.

**Q: What is the meeting slug?**
A: It is the last path segment of your HubSpot meeting link, stored in `hubspot_slug_name`.